CONFIDENTIAL DOCUMENT • COMMERCIAL & TECHNICAL PROPOSAL

MUTAMARK ENTERPRISES

Software Requirements Specification & Implementation Blueprint

Mutamark ERP — Integrated Enterprise Resource Planning & Paperless Operating System

DOCUMENT VERSION
1.0 (Master Proposal)

DATE
August 2026

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DOCUMENT STATUS
Approved Blueprint & Commercial Proposal

Prepared, owned, and architected by **WOLINET TECHNOLOGY & COOPERATION LTD** for Mutamark Enterprises.
Contains complete functional requirements, technical specifications, structured 3-week accelerated delivery timeline (21 days), and commercial pricing.

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional, technical, operational, and security requirements for the Mutamark ERP System. The system shall serve as the centralized digital operating platform for Mutamark Enterprises, integrating sales, procurement, inventory, accounting, human resources, projects, assets, document management, digital signatures, and business intelligence into a single unified environment.

> *"Enter information once, use it throughout the business."*

1.2 Document Conventions

TERM	DEFINITION
Shall	Mandatory requirement
Should	Recommended requirement
May	Optional requirement
TBD	To be determined during design phase

1.3 Intended Audience

This document is intended for system architects, database designers, software developers, quality assurance engineers, project managers, and Mutamark Enterprises stakeholders.

1.4 Project Scope

The Mutamark ERP shall replace fragmented business processes with integrated, paperless, audit-ready workflows. The system shall support multi-branch operations, role-based access, digital approvals, automated accounting entries, and real-time management dashboards.

1.5 References

- Mutamark Enterprises Business Requirements (Source Document)
- Applicable Tanzanian tax and e-invoicing regulations
- IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications

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2 Overall Description

2.1 Product Perspective

The ERP shall be a web-based, responsive application accessible via desktop, tablet, and mobile devices. It shall operate as a cloud-ready system with Progressive Web App (PWA) capabilities and offline synchronization for critical POS and

inventory operations.

2.2 Product Functions — Summary

The system shall provide the following functional domains:

1. System Administration & Security
2. Multi-Company & Branch Management
3. User, Role & Permission Management
4. CRM & Customer Management
5. Supplier & Vendor Management
6. Product & Service Catalog
7. Inventory & Warehouse Management
8. Procurement Management (Procure-to-Pay)
9. Sales Management (Quote-to-Cash)
10. Point of Sale (POS)
11. Accounting & Financial Management
12. Tax Management
13. Expense & Income Management
14. Cash & Bank Management
15. Human Resource Management
16. Attendance & Leave Management
17. Payroll Management
18. Project Management
19. Task Management
20. Fixed Asset Management
21. Warranty & Servicing
22. Document Management System (DMS)
23. Digital Signature & Verification
24. Approval Workflow Engine
25. Paperless Office & Template Engine
26. Notification & Communication Engine
27. Customer Self-Service Portal
28. Supplier Self-Service Portal
29. Barcode, QR & Serial Management
30. Reporting & Business Intelligence
31. Management Dashboard
32. API & External Integration
33. Payment Integration

34. Backup & Disaster Recovery

35. Audit Trail & Compliance

2.3 User Classes and Characteristics

USER CLASS	DESCRIPTION	PRIVILEGE LEVEL
Super Administrator	Full system control	Highest
Managing Director	Strategic oversight, all approvals	High
General Manager	Operational oversight	High
Finance Manager	Financial control and approvals	High
Accountant	Accounting entries and reconciliation	Medium-High
Procurement Manager	Procurement oversight and approvals	Medium-High
Procurement Officer	Purchase requisitions and orders	Medium
Sales Manager	Sales oversight and approvals	Medium-High
Sales Officer	Quotations and sales orders	Medium
Warehouse Manager	Inventory control and transfers	Medium
Storekeeper	Stock receipt and issue	Medium
Cashier	POS operations and cash handling	Low-Medium
HR Manager	Employee and payroll management	Medium-High
Project Manager	Project execution and costing	Medium
Auditor	Read-only access to all transactions	Medium
Customer	Self-service portal access	Low
Supplier	Self-service portal access	Low

2.4 Operating Environment

COMPONENT	SPECIFICATION
Server	Cloud or on-premise Linux/Windows server
Database	Relational database with ACID compliance
Client	Modern web browsers (Chrome, Firefox, Safari, Edge)
Mobile	iOS Safari, Android Chrome via PWA
Network	Internet connectivity with offline sync for POS

2.5 Design and Implementation Constraints

- The system shall support English as the primary interface language and shall be architected for future Kiswahili localization.
- Tax configuration shall be adaptable to Tanzanian tax requirements, including future e-invoicing (e-TIMS) integration.
- The system shall use double-entry accounting principles for all financial transactions.
- Document numbering shall be configurable by branch and financial year.

2.6 Assumptions and Dependencies

- Users shall have access to internet-connected devices.
- Barcode scanners, receipt printers, and cash drawers shall support standard connectivity protocols.
- Email, SMS, and WhatsApp gateway services shall be available for communication modules.

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3 System Features

3.1 MODULE 1 — SYSTEM ADMINISTRATION

3.1.1 The system shall allow authorized administrators to configure: Company profile, branding, and letterhead; Financial years and periods; Base currency and additional foreign currencies with exchange rates; Tax rates, tax groups, and tax-inclusive/exclusive pricing rules; Payment methods and terms; Document numbering sequences and formats; Branches, warehouses, and departments; Units of measure and product categories; System preferences and localization settings; Backup schedules and retention policies.

3.1.2 The system shall maintain an immutable audit log of all administrative configuration changes.

3.2 MODULE 2 — COMPANY & BRANCH MANAGEMENT

3.2.1 The system shall support: Management of one or more legal business entities; Multiple branches, stores, and warehouses per business entity; Separate stock balances, sales, and purchase tracking by location; Branch-specific user assignments and permissions; Branch-specific financial reporting; Consolidated group-level financial and operational reporting.

3.3 MODULE 3 — USER, ROLE & PERMISSION MANAGEMENT

3.3.1 The system shall implement Role-Based Access Control (RBAC) with configurable permissions for: View, Create, Edit, Delete, Approve, Reject, Print, Export, Download, Send, Sign, Pay, Cancel, Reverse, Import.

3.3.2 Permissions shall be granularly configurable by: Module, branch, warehouse, department, transaction type, and transaction amount threshold.

3.3.3 The system shall support: Unlimited user accounts; Custom role creation; Two-factor authentication (2FA); Strong password enforcement; Session timeout and concurrent session management; Login history and failed-login protection.

3.4 MODULE 4 — CRM & CUSTOMER MANAGEMENT

3.4.1 The system shall maintain a Customer Master with: Customer ID, name, company name, contact person; Phone, email, physical address, tax identification; Customer category, type (individual/company), assigned salesperson; Credit limit, payment terms, default discount, opening balance; Attached documents and notes.

3.4.2 The system shall provide: Customer registration and history tracking; Customer grouping and categorization; Credit limit enforcement with override controls; Customer statement generation (transactional and outstanding balance); Customer communication history logging; Customer enquiry and follow-up tracking; Complaint and support ticket management.

3.5 MODULE 5 — SUPPLIER & VENDOR MANAGEMENT

3.5.1 The system shall maintain a Supplier Master with: Supplier ID, name, contact person, phone, email, address; Tax/VAT information, bank details, payment terms; Supplier category, opening balance, performance rating; Attached documents and qualification records.

3.5.2 The system shall provide: Supplier ledger with full transaction history; Purchase order, GRN, and invoice linkage; Supplier payment tracking and advance management; Supplier aging and outstanding balance reports; Supplier performance evaluation and rating.

3.6 MODULE 6 — PRODUCT & SERVICE CATALOG

3.6.1 The system shall support the following product types: General products; Variable products (with attributes); Services; Serial-number products; IMEI-tracked products; Batch/lot-tracked products; Expiry-date products; Installment products; Combo/kit products.

3.6.2 The Product Master shall contain: Product ID, SKU (auto-generated), barcode, product name, alternative name; Description, brand, category, sub-category, unit of measure; Purchase price, selling price, wholesale price, minimum price; Configurable profit margin with automatic selling-price calculation; Tax assignment, reorder level, maximum stock, preferred supplier; Default warehouse, product image; Serial number, batch number, and expiry date tracking where applicable; Warranty period.

3.7 MODULE 7 — INVENTORY & WAREHOUSE MANAGEMENT

3.7.1 The system shall maintain real-time stock balances across all warehouses with full traceability.

3.7.2 The system shall support the following stock transaction types: Opening stock; Purchase receipt; Sales issue; Sales return; Purchase return; Warehouse-to-warehouse transfer; Stock adjustment (positive/negative); Stock count/physical verification; Damaged stock write-off; Expired stock write-off; Stock issue for projects; Stock receipt from returns.

3.7.3 The system shall provide: Live stock status and valuation (FIFO, Average Cost, or Standard Cost); Low-stock and reorder-level alerts; Stock movement history and ledger; Damaged, expired, and lost stock tracking; Barcode scanning for stock operations; Negative stock prevention (configurable by product/warehouse).

3.8 MODULE 8 — PROCUREMENT MANAGEMENT (PROCURE-TO-PAY)

3.8.1 The system shall enforce the following workflow:

```
Purchase Requisition → Approval → RFQ → Supplier Quotations
→ Comparative Evaluation → Purchase Order → GRN
→ Supplier Invoice → Three-Way Matching → Payment → Supplier Statement
```

3.8.2 **Purchase Requisition:** Requisition number, department, requester, date, required date; Items, quantities, estimated prices, purpose, budget reference; Attachment support and multi-level approval routing.

3.8.3 **Request for Quotation (RFQ):** Supplier selection, RFQ issuance, quotation receipt; Price, delivery period, and payment term recording; Automatic comparative statement generation.

3.8.4 **Purchase Order (PO):** PO number, supplier, items, quantities, prices, discounts, taxes; Delivery terms, payment terms, delivery location; Authorized signature placeholders; Automatic inventory commitment upon approval.

3.8.5 **Goods Received Note (GRN):** GRN creation against PO with quantity variance tracking; Automatic inventory update upon GRN posting; Quality inspection status and rejection handling.

3.8.6 **Three-Way Matching:** The system shall enforce matching of PO, GRN, and Supplier Invoice before payment authorization, with configurable tolerance for quantity and price variances.

3.8.7 **Supplier Invoice & Payment:** Supplier invoice recording and linkage to PO/GRN; Payment voucher generation, advance payment tracking; Supplier payment scheduling and due-date alerts.

3.8.8 The system shall provide a procurement dashboard with pending requisitions, open POs, overdue deliveries, and purchase budget utilization.

3.9 MODULE 9 — SALES MANAGEMENT (QUOTE-TO-CASH)

3.9.1 The system shall enforce the following workflow:

```
Customer Enquiry → Quotation → Customer Approval → Sales Order
→ Stock Allocation → Delivery Note → Invoice → Receipt/Payment
→ Customer Statement
```

3.9.2 **Quotation:** Creation, editing, approval, rejection, expiry date management; Email, PDF download, print, and WhatsApp sharing; Digital signature application; Conversion to Sales Order or Invoice.

3.9.3 **Proforma Invoice:** Proforma generation, sharing, and conversion to formal invoice.

3.9.4 **Sales Order:** Customer confirmation tracking, stock reservation; Partial and full fulfillment management.

3.9.5 **Delivery Note:** Delivery creation from sales order or invoice; Partial delivery support, delivery confirmation; Serial/batch number capture at point of delivery.

3.9.6 **Invoice:** Automatic invoice generation from delivery or sales order; Tax calculation, discount application, credit limit validation; Automatic accounting entry generation (Dr. AR, Cr. Revenue, Cr. Tax); PDF generation with QR code for verification; Email, SMS, WhatsApp distribution.

3.9.7 **Receipt & Payment:** Payment allocation to specific invoices; Advance receipt handling, overpayment management; Automatic accounting entry (Dr. Bank/Cash, Cr. AR).

3.9.8 **Credit & Debit Notes:** Credit note issuance for returns or adjustments; Debit note issuance for additional charges.

3.9.9 The system shall provide sales return processing with automatic inventory and accounting reversal.

3.10 MODULE 10 — POINT OF SALE (POS)

3.10.1 The POS module shall support: Cash sales, credit sales, and multiple payment methods per transaction; Barcode scanning, product search, and quick product selection; Customer selection, discount application, and tax calculation; Held/parked sales and retrieval; Sales returns and refund processing; Thermal receipt printing, customer-facing display; Cash drawer management and POS register balancing; Daily closing and Z-report generation; Cash reconciliation and variance reporting.

3.11 MODULE 11 — ACCOUNTING & FINANCE

3.11.1 The system shall implement double-entry accounting integrated with all commercial transactions.

3.11.2 **Chart of Accounts:** Hierarchical chart with categories: Assets, Liabilities, Equity, Revenue, Cost of Sales, Expenses; Account codes, names, types, and parent-child relationships; Opening balance entry and financial year closing.

3.11.3 **Core Accounting Functions:** General Ledger with full transaction drill-down; Journal entries with approval workflow; Cash book and bank account management; Accounts Receivable and Accounts Payable sub-ledgers; Trial Balance, Profit & Loss Statement, Balance Sheet, Cash Flow Statement; Bank reconciliation with statement import; Opening and closing balance management.

3.11.4 **Automated Accounting Entries:** The system shall automatically generate accounting entries for: Sales invoices (Dr. AR, Cr. Revenue, Cr. Tax Payable); Customer payments (Dr. Bank/Cash, Cr. AR); Purchase receipts (Dr. Inventory, Cr. AP); Supplier payments (Dr. AP, Cr. Bank/Cash); Stock adjustments (Dr./Cr. Inventory, Cr./Dr. Adjustment Account); Expense payments (Dr. Expense, Cr. Bank/Cash); Payroll posting (Dr. Salary Expense, Cr. Bank/Cash, Cr. Deductions Payable).

3.12 MODULE 12 — TAX MANAGEMENT

3.12.1 The system shall support: Configurable tax rates and tax groups; Tax-inclusive and tax-exclusive pricing models; Product-specific and transaction-specific tax assignment; Sales tax, purchase tax, withholding tax tracking; Tax liability reports and tax return support; Future-ready architecture for Tanzanian e-invoicing (e-TIMS) integration.

3.13 MODULE 13 — EXPENSE & INCOME MANAGEMENT

3.13.1 The system shall support configurable expense categories including: Rent, utilities, transport, communication, salaries, office expenses, repairs, marketing, bank charges, taxes, and other expenses.

3.13.2 Each expense shall require: Expense request, approval workflow, payment voucher, and accounting entry.

3.13.3 The system shall track non-sales income sources with categorization and reporting.

3.14 MODULE 14 — CASH & BANK MANAGEMENT

3.14.1 The system shall manage: Multiple cash accounts and bank accounts; Deposits, withdrawals, and inter-account transfers; Bank reconciliation with unreconciled transaction tracking; Cash reconciliation and float management; Payment vouchers and receipt vouchers; Bank statement import (CSV, Excel, MT940 where applicable).

3.15 MODULE 15 — HUMAN RESOURCE MANAGEMENT

3.15.1 The system shall maintain an Employee Master with: Employee ID, personal details, employment details; Department, position, employment status, contract details; Salary structure, allowances, deductions; Attached documents (contracts, certificates, IDs).

3.15.2 The system shall provide: Employee directory and organizational chart; Contract management and renewal alerts; Employee performance tracking; Commission and incentive calculation; Employee exit/termination management.

3.16 MODULE 16 — ATTENDANCE & LEAVE MANAGEMENT

3.16.1 The system shall support: Daily check-in/check-out with timestamp; Attendance correction requests and approval; Late arrival, early departure, and absence flagging; Leave type configuration (annual, sick, maternity, etc.); Leave application, approval workflow, and balance tracking; Holiday calendar management; Attendance reports and payroll integration.

3.17 MODULE 17 — PAYROLL MANAGEMENT

3.17.1 The system shall calculate: Basic salary, allowances, deductions, overtime; Commissions, loans, and salary advances; Statutory deductions (PAYE, NSSF, NHIF, WCF as applicable); Net salary.

3.17.2 The system shall generate: Payroll register and summary; Individual payslips (printable and downloadable); Salary history and variance reports; Automatic accounting posting for payroll.

3.18 MODULE 18 — PROJECT MANAGEMENT

3.18.1 The system shall support: Project registration with unique project number; Customer assignment, project manager assignment; Start date, end date, budget allocation; Actual expenditure tracking (materials, labor, expenses); Revenue recognition and profitability calculation; Milestone definition and tracking; Project document repository; Customer portal viewing and comment functionality.

3.19 MODULE 19 — TASK MANAGEMENT

3.19.1 The system shall provide: Task creation, assignment, and deadline setting; Priority levels and status tracking: Draft → Assigned → In Progress → Review → Completed → Closed; Comment threads and file attachments; Calendar view and milestone linkage; Completion percentage and time logging.

3.20 MODULE 20 — FIXED ASSET MANAGEMENT

3.20.1 The system shall manage: Asset registration with unique asset code; Asset category, purchase cost, purchase date, location; Responsible employee/custodian, supplier, warranty details; Depreciation calculation (straight-line, reducing balance, or custom methods); Asset transfer between locations or employees; Asset disposal and write-off; Physical verification tracking and asset history log.

3.21 MODULE 21 — WARRANTY & SERVICING

3.21.1 The system shall support: Warranty registration linked to product serial numbers and invoices; Warranty period tracking and expiry alerts; Customer warranty lookup by invoice, customer name, or phone number; Service request intake and repair ticket management; Vendor repair workflow (send to vendor, track status, receive back); Paid servicing billing and customer notification; Complete service history per product/serial number.

3.22 MODULE 22 — DOCUMENT MANAGEMENT SYSTEM (DMS)

3.22.1 The system shall provide a central electronic document repository attaching files to: Customers, suppliers, employees, products; Purchases, sales, projects, assets, payments, contracts.

3.22.2 Supported formats shall include PDF, Word, Excel, images, and scanned documents.

3.22.3 The system shall maintain document version history with upload timestamp and user attribution.

3.23 MODULE 23 — DIGITAL SIGNATURE & VERIFICATION

3.23.1 The system shall integrate digital signatures into document workflows.

3.23.2 Each digital signature shall record: Signatory name and position; Date and timestamp; Document ID and version; Signature status and verification identifier; IP address and device information where appropriate.

3.23.3 Signed documents shall display: Digital approval block with signatory details; Verification ID and QR code for third-party verification.

3.23.4 The system shall provide a public verification page where any party can verify document authenticity using the document number and verification code.

3.24 MODULE 24 — APPROVAL WORKFLOW ENGINE

3.24.1 The system shall provide a configurable workflow engine with statuses: Draft → Submitted → Under Review → Returned → Approved → Rejected → Signed → Issued → Closed.

3.24.2 Approval rules shall support: User-based, role-based, and department-based routing; Amount-based escalation (e.g., ≤500,000 Procurement; 500,001–5,000,000 Manager; >5,000,000 Director); Sequential and parallel approval paths; Delegated approval during absence; Segregation of duties (a user shall not approve their own transactions where prohibited).

3.24.3 All approval actions shall be recorded in the audit trail with before/after values.

3.25 MODULE 25 — PAPERLESS OFFICE & TEMPLATE ENGINE

3.25.1 The system shall minimize physical document dependency through digital workflow: Prepare → Review → Approve → Digitally Sign → Generate PDF → Send → Archive.

3.25.2 The template engine shall support dynamic field substitution for: Company details, customer/supplier details, document numbers, dates; Line items, subtotals, taxes, grand totals, amounts in words; Prepared by, approved by, signature blocks.

3.25.3 Templates shall be available for: Quotation, Proforma Invoice, Sales Order, Delivery Note, Invoice, Receipt; Purchase Requisition, RFQ, Comparative Statement, Purchase Order, GRN; Credit Note, Debit Note, Payment Voucher, Receipt Voucher, Journal Voucher; Customer Statement, Supplier Statement, Payslip, Contracts.

3.26 MODULE 26 — NOTIFICATION & COMMUNICATION ENGINE

3.26.1 The system shall generate automatic alerts for: Low stock, reorder levels, stock expiry; Overdue customer invoices and supplier payments; Quotation expiry, contract expiry, warranty expiry; Pending approvals, pending deliveries, pending purchase orders; Employee leave requests, payroll processing reminders; Task and project deadline reminders.

3.26.2 Notification channels shall include: In-system notification center; Email, SMS, WhatsApp messaging; Mobile push notifications (via PWA).

3.27 MODULE 27 — CUSTOMER SELF-SERVICE PORTAL

3.27.1 Customers shall be able to: Securely log in with username/password or token-based access; View quotations, approve quotations, and request revisions; View sales orders, delivery notes, and invoices; Download documents and view account statements; Make online payments and track payment history; Submit support tickets and track order status.

3.28 MODULE 28 — SUPPLIER SELF-SERVICE PORTAL

3.28.1 Suppliers shall be able to: Receive RFQs and submit quotations; Receive purchase orders and confirm acceptance; Upload invoices and delivery notes; View payment status and purchase history; Update company information and contact details.

3.29 MODULE 29 — BARCODE, QR & SERIAL MANAGEMENT

3.29.1 The system shall support: Barcode generation and label printing (standard formats: Code 128, EAN, UPC); QR code generation for document verification and product tracking; Barcode scanning for stock receipt, stock issue, and POS operations; Serial number and IMEI scanning and validation; Stock-taking via barcode scanner.

3.30 MODULE 30 — REPORTING & BUSINESS INTELLIGENCE

3.30.1 **Sales Reports:** Daily, monthly, and period sales reports; Sales by customer, product, branch, salesperson; Sales return analysis and salesperson performance.

3.30.2 **Purchase Reports:** Purchase summary, supplier purchase history; Product purchase analysis, purchase returns; Supplier performance and price variance.

3.30.3 **Inventory Reports:** Stock balance, stock movement, stock valuation; Low stock, damaged stock, expired stock; Warehouse-wise stock and transfer reports.

3.30.4 **Financial Reports:** Trial Balance, General Ledger, Profit & Loss, Balance Sheet, Cash Flow; Accounts Receivable aging, Accounts Payable aging; Expense analysis and budget vs. actual.

3.30.5 **HR Reports:** Employee directory, attendance summary, leave balance; Payroll register, salary history, commission reports.

3.30.6 All reports shall support PDF export, Excel/CSV download, and date range filtering.

3.31 MODULE 31 — MANAGEMENT DASHBOARD

3.31.1 The dashboard shall display real-time KPIs including:

CATEGORY	KPIS
Financial	Revenue, Gross Profit, Net Profit, Expenses, Cash/Bank Balance, Receivables, Payables
Operational	Sales count, Purchase value, Stock value, Low-stock items, Pending orders/approvals/deliveries
HR	Employee count, attendance status, pending leave, monthly payroll
Project	Active projects, budget utilization, actual cost, profitability

3.31.2 The dashboard shall be role-aware, displaying only relevant metrics based on user permissions.

3.32 MODULE 32 — API & EXTERNAL INTEGRATION

3.32.1 The system shall provide a secure REST API with: OAuth 2.0 or API key authentication; Endpoints for customers, suppliers, products, inventory, sales, purchases, invoices, payments, accounting, employees; Webhook support for event-driven integrations; Rate limiting and request logging.

3.32.2 The system shall support data import/export via Excel and CSV templates.

3.33 MODULE 33 — PAYMENT INTEGRATION

3.33.1 The system shall support configurable payment methods: Cash, Bank transfer, Card, Mobile money (M-Pesa, Tigo Pesa, Airtel Money); Online payment gateways; Payment links for customer portal invoices.

3.33.2 Each payment shall be automatically linked to the originating invoice, customer/supplier account, and cash/bank ledger.

3.34 MODULE 34 — BACKUP & DISASTER RECOVERY

3.34.1 The system shall provide: Automated daily incremental backups and periodic full backups; Manual backup initiation; Backup encryption and off-site storage; Point-in-time restore capability; Backup history and integrity verification; Documented disaster recovery procedures.

3.35 MODULE 35 — AUDIT TRAIL & COMPLIANCE

3.35.1 The system shall record every sensitive transaction with: User identity, action type, date/time, IP address; Document number, previous value, new value; Approval status change.

3.35.2 Audit records shall be immutable and non-deletable through standard application functions.

3.35.3 The system shall provide audit trail filtering and export for compliance and forensic review.

4 External Interface Requirements

4.1 User Interfaces

- Responsive web interface optimized for 1920×1080, 1366×768, and mobile viewports
- Consistent navigation, form validation, and error messaging
- Accessibility compliance (WCAG 2.1 Level AA where practicable)

4.2 Hardware Interfaces

DEVICE	INTERFACE
Barcode scanner	USB HID and serial
Thermal receipt printer	ESC/POS protocol
Cash drawer	Printer-driven or USB
Customer pole display	Serial/USB
Weighing scale	RS232/USB where applicable

4.3 Software Interfaces

- Email server (SMTP)
- SMS gateway (REST API)
- WhatsApp Business API
- Payment gateway APIs
- Future e-TIMS tax integration API

4.4 Communication Interfaces

- HTTPS/TLS 1.3 for all data transmission
- REST API over JSON
- WebSocket for real-time notifications where applicable

5 Non-Functional Requirements

5.1 Performance Requirements

- Page load time shall not exceed 3 seconds under normal network conditions
- The system shall support a minimum of 50 concurrent users per branch
- Database queries shall be optimized with appropriate indexing
- Result sets shall implement server-side pagination

5.2 Security Requirements

- Passwords shall be hashed using bcrypt or equivalent
- All sensitive data shall be encrypted at rest and in transit
- API endpoints shall require authenticated access tokens
- File uploads shall be scanned and restricted to permitted MIME types
- SQL injection, XSS, and CSRF protection shall be implemented

5.3 Availability and Reliability

- Target uptime of 99.5% during business hours
- Automated failover for critical database operations
- Graceful degradation of non-critical services

5.4 Scalability

- The architecture shall support horizontal scaling for additional branches, warehouses, and users without fundamental redesign
- Database partitioning strategy shall accommodate 10+ years of transaction history

5.5 Usability

- Interface shall be intuitive for non-technical users
- Contextual help and field-level tooltips
- Consistent color coding for status (Draft, Approved, Rejected, Paid, etc.)

5.6 Maintainability

- Modular codebase enabling independent module updates
- Comprehensive API documentation and code commenting
- Environment-specific configuration management

5.7 Portability

- Browser-based access eliminating client installation
- PWA enabling mobile home-screen installation
- Database abstraction layer supporting multiple RDBMS engines

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6 Data Requirements

6.1 Data Model Overview

The database shall be normalized to 3NF for master data with appropriate denormalization for reporting tables. All financial transactions shall maintain referential integrity.

6.2 Major Entity Groups

DOMAIN	KEY ENTITIES
Organization	Company, Branch, Department, Warehouse, Financial Year
Security	User, Role, Permission, UserRole, AuditLog, Session
CRM	Customer, CustomerContact, CustomerGroup, CustomerTransaction
Procurement	PurchaseRequisition, RFQ, SupplierQuotation, PurchaseOrder, GRN, SupplierInvoice, SupplierPayment
Sales	Enquiry, Quotation, SalesOrder, DeliveryNote, Invoice, Receipt, CreditNote, SalesReturn
Inventory	Product, Category, Brand, Unit, Warehouse, Stock, StockMovement, StockTransfer, SerialNumber, Batch
Accounting	Account, Journal, JournalEntry, Payment, Receipt, BankAccount, Tax
HR	Employee, Department, Position, Attendance, Leave, Payroll, Payslip
Assets	Asset, AssetCategory, Depreciation, AssetTransfer, AssetDisposal
Documents	Document, DocumentTemplate, DocumentVersion, DigitalSignature, Approval, Attachment
Projects	Project, Task, Milestone, ProjectCost

6.3 Data Retention

DATA TYPE	RETENTION PERIOD
Active transaction data	Indefinite
Audit logs	Minimum 7 years
Deleted records	Soft delete with 90-day grace period, then archive

6.4 Backup and Recovery

See Module 34.

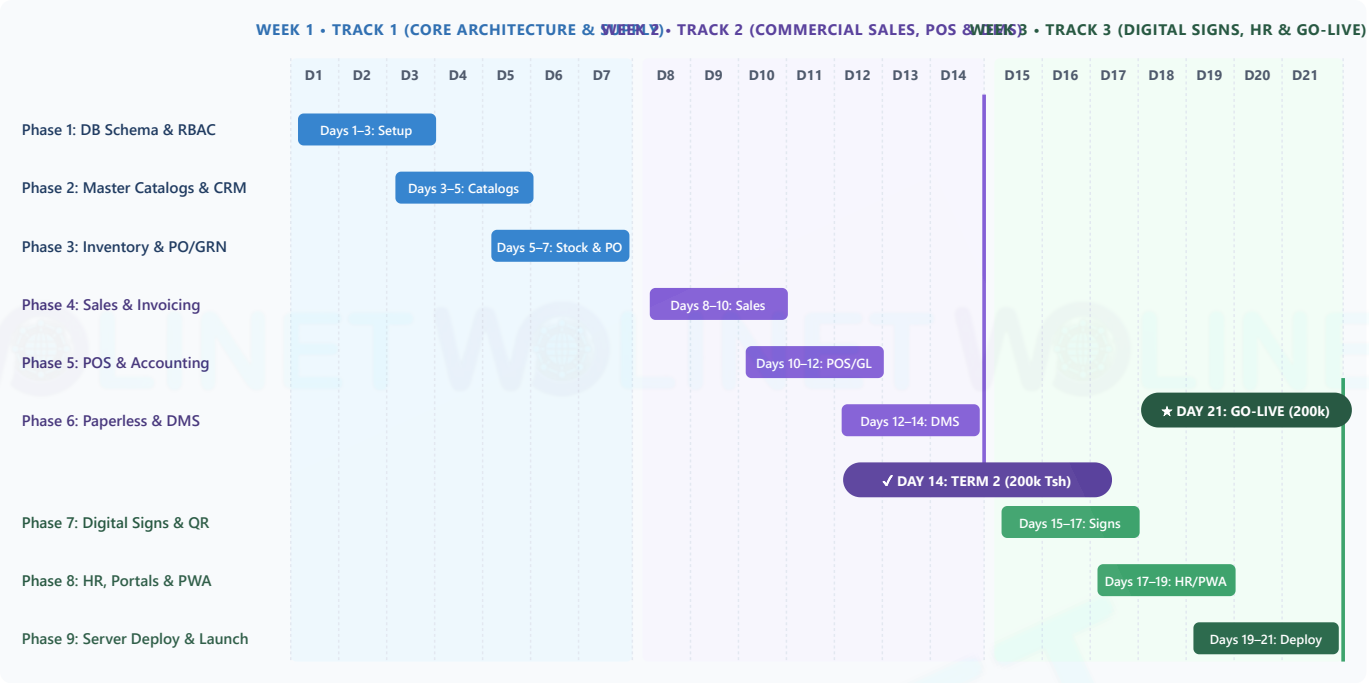
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7 Development Roadmap & Three-Week Delivery Timeline (21 Days)

To ensure structured implementation, immediate operational value, and zero business downtime, the entire implementation of the **Mutamark ERP System** is engineered into a **Structured 3-Week Sprint Execution Model (21 Calendar Days Total)** divided into three sequential weekly project tracks (Week 1, Week 2 & Week 3). Each 1-week track delivers an autonomous, fully functional milestone synchronized with the 3-term payment schedule (200,000 TZS per term).

Executive Project Timeline & 21-Day Delivery Gantt Chart

Structured 3-Week Turnkey Delivery (21 Days)



PROJECT TRACK 1 • WEEK 1

Days 1 – 7 (Week 1)

Core Architecture & Supply Chain

- **Days 1–2 (Architecture & Schema):** Multi-entity database tables, RBAC, branch & warehouse structures.
- **Days 2–4 (Master Data & CRM):** Product catalogs, variants, units, customer/supplier ledgers & opening balances.
- **Days 4–7 (Supply Chain & Stock):** Real-time inventory tracking, multi-warehouse transfers, PO, GRN & three-way match.

🚀 **Term 1 Deliverable:** Project Kickoff & Core Supply Chain Staging Demo (Term 1 Deposit: 200,000 TZS)

PROJECT TRACK 2 • WEEK 2

Days 8 – 14 (Week 2)

Sales, POS & Paperless DMS

- **Days 8–10 (Sales & Financials):** Quotations, proforma invoices, sales orders, delivery notes, invoices & receipts.
- **Days 10–12 (POS Counter & Accounting):** Rapid POS counter, receipt printing, double-entry auto-journal posting & GL.
- **Days 12–14 (Paperless Office DMS):** Dynamic document template builder, PDF generator engine & multi-tier approvals.

Term 2 Deliverable: Mid-Term Commercial Operations & DMS Demo (Triggers Term 2: 200,000 TZS)

PROJECT TRACK 3 • WEEK 3

Days 15 – 21 (Week 3)

Digital Trust, HR & Go-Live

- **Days 15–17 (Digital Signatures & QR):** Cryptographic signing, tamper-evident seals & public QR verification portal.
- **Days 17–19 (HR, Payroll & Portals):** Employee directory, attendance, payroll posting, assets, customer/supplier portals & PWA.
- **Days 19–21 (Production Deploy & Launch):** VPS server provisioning, SSL certificate, automated daily DB backups & Go-Live.

✨ **Term 3 Deliverable:** Full Production Handover + 1-Yr Free Support (Triggers Final Term 3: 200,000 TZS)

8 Commercial Investment, Pricing & Payment Schedule

Wolinet Technology & Cooperation Ltd is pleased to submit this competitive commercial investment proposal for the turnkey implementation, deployment, and ongoing maintenance of the Mutamark ERP System. Our pricing structure provides maximum return on investment for Mutamark Enterprises with transparent pricing, guaranteed delivery, and generous partnership discounts.

★ SPECIAL STRATEGIC PARTNERSHIP • 60% DISCOUNT APPLIED

Complete Mutamark ERP System Package

Comprehensive enterprise software package covering all 35 integrated modules, cryptographic digital signatures, paperless office engine, production server deployment, and 1 full year of 24/7 technical support.

- ✓ Zero Recurring License Fees
- ✓ 100% Full System Ownership
- ✓ Free Server Deployment

Standard System Valuation	~~1,500,000 TZS~~
Strategic Partner Discount	-60% DISCOUNT
Discount Amount Saved	-900,000 TZS

TOTAL NET PAYABLE

600,000 TZS

Six Hundred Thousand Tanzanian Shillings Only (TSH 600,000/=)

8.1 Detailed Scope of Investment

#	MODULE SCOPE & DELIVERABLES	STANDARD VALUATION	DISCOUNTED PRICE
1	**Core ERP Engine & 35 Operational Modules** Inventory, Procurement, Sales, POS, Accounting, Multi-Branch, HR, Assets, CRM, Reporting	1,000,000 TZS	**350,000 TZS**
2	**Paperless Office DMS, Dynamic Template Builder & Digital Signatures** Cryptographic digital signature verification, multi-level approval workflows, QR code verification	300,000 TZS	**250,000 TZS**
3	**Production System Server Deployment & Environment Configuration** Linux/Windows server setup, Nginx/Apache configuration, SSL certificate, automated daily database backup	~~200,000 TZS~~	**100% FREE**

#	MODULE SCOPE & DELIVERABLES	STANDARD VALUATION	DISCOUNTED PRICE
4	**1 Year Full Technical Support, Maintenance & Health Monitoring** 24/7 SLA incident resolution, software updates, security patches, database index maintenance	~~350,000 TZS~~	**100% FREE**
	Total Gross Valuation:	~~**1,500,000 TZS**~~	—
	Special Strategic Discount (60% OFF):	**-900,000 TZS**	
	Net Payable Investment:	**TSH 600,000/= (600k Tsh)**	

8.2 Milestone Payment Schedule (3 Terms • 3-Week Delivery Plan)

Payments are structured into **three equal installments (terms) of 200,000 TZS each (Total TSH 600,000/=)** strictly aligned with the 3-week delivery milestones:

TERM 1 • DAY 1 KICKOFF

INITIAL DEPOSIT / TERM 1 (33.3%)

200,000 TZS

Payable upon Project Kickoff (Day 1 / Term 1)
Triggers immediate architectural provisioning, database schema setup, multi-branch setup, and execution of Week 1 Track.

TERM 2 • DAY 14 MILESTONE

MID-PROJECT SIGN-OFF / TERM 2 (33.3%)

200,000 TZS

Payable upon Week 2 Demo (Day 14 / Term 2)
Due upon successful demonstration of Commercial Operations (Sales, Invoicing, Receipts, POS counter, Accounting integration & Paperless DMS workflows).

TERM 3 • DAY 21 GO-LIVE

FINAL GO-LIVE & HANDOVER / TERM 3 (33.3%)

200,000 TZS

Payable upon Full Go-Live (Day 21 / Term 3)
Due upon production server deployment, SSL activation, digital signatures & QR verification sign-off, and final handover.

8.3 Official Payment & Banking Details (Lipa Namba & Scan to Pay)

Clients and stakeholders can disburse project milestone payments securely via our unified merchant payment channel supporting all Tanzanian Mobile Money operators and Commercial Banks:

💳 VERIFIED COMMERCIAL PAYMENT CHANNEL

Wolinet Technology & Cooperation Ltd

Use the official merchant **Lipa Namba / Account Number** below for all installment deposits (Term 1, Term 2 & Term 3):

LIPA NAMBA / ACCOUNT NO.

16381432

ACCOUNT / MERCHANT NAME

WOLINET

📱 **Mobile Money (Mitandao Yote):** M-Pesa, Tigo Pesa, Airtel Money, Halopesa
→ Piga menu ya Lipa kwa Simu → Chagua Lipa kwa Namba → Weka **16381432** → Thibitisha Jina **WOLINET**

🏦 **Benki Zote (All Banks):** CRDB SimBanking, NMB Mkononi, Stanbic, KCB, NBC, DTB, Standard Chartered, etc.
→ Fungua App ya Benki → Chagua Pay Merchant / Lipa Namba → Weka **16381432** → Thibitisha **WOLINET**

📝 **Reference / Kumbukumbu:** Mutamark ERP — Term Deposit

LIPA KWA KUSCAN
Instant QR Code Payment



Lipa Namba: 16381432
Lipa kutoka mitandao yote na Benki

8.4 Special Value-Added Inclusions (100% Free of Charge)

★ 1 Year Free Technical Support & SLA — 100% FREE

Mutamark Enterprises receives a full **12-month complimentary technical maintenance package** backed by dedicated engineers:

- 24/7 SLA emergency assistance and rapid bug remediation
- Routine database maintenance, health checks & query optimization
- Security patches, framework updates & performance tuning
- Dedicated remote helpdesk support for administrators

🚀 Free Production Server Deployment — 100% FREE

Zero server setup fees — Wolinet completely handles end-to-end **production deployment & infrastructure hardening**:

- Full cloud VPS / dedicated server environment provisioning
- Nginx / Apache web server & application runtime configuration
- Automated daily database backup scripts & disaster recovery
- Free SSL/TLS security encryption certificate & custom domain binding

9 Acceptance Criteria

The ERP shall be considered ready for production when the following criteria are satisfied across all three project tracks:

1. Authorized users can securely authenticate and access permitted functions.
2. Role-based permissions correctly restrict access by module, branch, and transaction amount.
3. Products, customers, and suppliers can be fully managed with complete history.
4. Purchase transactions update inventory and accounts payable automatically.
5. Sales transactions update inventory, accounts receivable, and revenue automatically.
6. Customer and supplier payments correctly update balances and bank/cash ledgers.
7. Financial reports reconcile to the transaction level with zero unexplained variances.
8. Approval workflows route correctly through configured hierarchies.
9. Digital signatures can be applied, displayed, and independently verified.
10. Documents can be generated as PDF, distributed electronically, and archived.
11. Audit logs capture all sensitive actions with before/after values.
12. Reports can be generated, filtered, and exported in PDF and Excel formats.
13. Data can be backed up, verified, and restored within documented RTO/RPO.
14. Unauthorized access attempts are prevented and logged.
15. Multiple branches operate independently while consolidated reporting reflects group position.
16. System performance remains acceptable under expected concurrent user load.
17. Critical workflows have been executed, server deployed, and signed off by Mutamark management.

10 Appendices

Appendix A — Master Document Numbering Scheme

DOCUMENT TYPE	FORMAT EXAMPLE
Quotation	QT-2026-00001
Proforma Invoice	PI-2026-00001
Sales Order	SO-2026-00001
Delivery Note	DN-2026-00001
Invoice	INV-2026-00001
Credit Note	CN-2026-00001
Purchase Requisition	PR-2026-00001

DOCUMENT TYPE	FORMAT EXAMPLE
Request for Quotation	RFQ-2026-00001
Purchase Order	PO-2026-00001
Goods Received Note	GRN-2026-00001
Payment Voucher	PV-2026-00001
Receipt Voucher	RV-2026-00001
Journal Voucher	JV-2026-00001

Appendix B — Core Business Controls

CONTROL AREA	RULE
Invoice Integrity	Finalized invoices shall not be editable; corrections require credit notes.
Payment Control	Payments shall not exceed outstanding balance without authorized override.
Stock Control	Negative stock shall be prohibited unless explicitly authorized per product.
Procurement Control	Supplier payments shall require verified PO, GRN, and Invoice (three-way match).
Accounting Control	Posted journal entries shall not be deleted; corrections require reversal entries.
Segregation of Duties	Users shall not approve their own transactions where separation is required.

Appendix C — Glossary

TERM	DEFINITION
ERP	Enterprise Resource Planning
GRN	Goods Received Note
PO	Purchase Order
RFQ	Request for Quotation
PWA	Progressive Web App
RBAC	Role-Based Access Control
SRS	Software Requirements Specification
TZS	Tanzanian Shilling
2FA	Two-Factor Authentication
DMS	Document Management System
AR	Accounts Receivable
AP	Accounts Payable
P&L	Profit and Loss Statement
KPI	Key Performance Indicator
BI	Business Intelligence
API	Application Programming Interface
RTO	Recovery Time Objective

TERM	DEFINITION
RPO	Recovery Point Objective

11 Recommended Next Steps & Commercial Sign-Off

Upon formal acceptance of this Software Requirements Specification and Commercial Proposal, the project will immediately commence with the following 3-week execution milestones:

1. **Day 1: Execution of Contract & Term 1 Upfront Deposit (200,000 TZS)** — Formal project authorization, multi-branch setup, and immediate database schema provisioning.
2. **Week 1 (Days 1–7): Core Architecture, Master Data & Supply Chain Foundation** — Table design, RBAC permissions, branch structure, master catalogs, real-time inventory, and procurement workflows.
3. **Week 2 (Days 8–14): Commercial Sales, Point of Sale (POS), Accounting & Paperless DMS** — Sales cycle, thermal POS counter, double-entry auto-journal links, document templates, and DMS.
4. **Day 14: Mid-Term Demonstration & Term 2 Milestone Payment (200,000 TZS)** — Live demonstration and formal sign-off of commercial operations and paperless workflows.
5. **Week 3 (Days 15–21): Digital Trust, HR Automation, Server Deployment & Go-Live** — Cryptographic digital signatures, QR verification, HR, payroll, customer portal, production VPS deployment, SSL security, and staff training.
6. **Day 21: Full Production Handover, Term 3 Final Payment (200,000 TZS) & 1-Year Free Support Kickoff** — Production launch, final handover, and 12-month complimentary technical support activation.

Payment Deposit Account / Lipa Namba:
Lipa Namba: **16381432** | Merchant: **WOLINET** (Mitandao yote & Benki zote)

Scan QR Code in Section 8.3

Immediate 3-Week Turnkey Delivery Guarantee

The development team at **WOLINET TECHNOLOGY & COOPERATION LTD** is prepared to commence immediately upon receipt of the initial Term 1 deposit (200,000 TZS), delivering the comprehensive commercial demo within **2 weeks (Day 14)** and full production go-live, server deployment & staff handover within **3 weeks (Day 21)**.

11.1 Document & Commercial Approval Authorization

Authorized on behalf of Mutamark Enterprises:

Authorized Signatory: _____

Designation / Title: _____

Signature: _____ Date: ____ / ____ / 2026

Prepared & Submitted by Wolinet:

Entity: WOLINET TECHNOLOGY & COOPERATION LTD

Contact: info@wolinet.com | +255 747 003 357

Authorized Signatory: Wolinet Ltd Date: August 2026

WOLINET TECHNOLOGY • ENTERPRISE SOLUTIONS

WOLINET TECHNOLOGY & COOPERATION LTD

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Document Version 1.0 (Master Commercial & Technical Blueprint) | August 2026